# Neoantimon2 — input and output examples

Independent illustrative snippets • Not a single reconstructed case or actual predictor output

## WHAT YOU PROVIDE

### Patients, samples and VCF

P1 → S1, S2; P2 → S3 (explicit mappings)
VCF columns: CHROM POS REF ALT FORMAT S1
Example row: chr1 4 G A GT 0/1
Each VCF has exactly one sample column.

### Patient-level HLA declarations

P1: HLA-A*02:01 / HLA-B*07:02
Class II presenter requires both chains:
HLA-DRA*01:01~HLA-DRB1*15:01
User supplied; no automatic HLA typing.

### Versioned reference resources

GRCh38.p14 genome + matching FAI
GENCODE 50 transcript / protein catalogs
MANE Select 1.5 summary / GTF
Explicit compatible files, not patient attributes.

### Optional supporting observations

RNA read / expression evidence
Cancer-cell fraction (CCF) / clonality inputs
Mass-spectrometry (MS) observations
Missing evidence is not a measured negative.

## WHAT YOU INSPECT

### Events and origins

Event: normalized chr1:4 G>A
Origin: transcript TA.1 + translation path
Origins retain contributing event links.
One event can participate in several paths.

### Peptide and presenter

Peptide: SIINFEKLA (illustrative sequence)
Candidate: SIINFEKLA + HLA-A*02:01
Distinct origins may share this peptide.
Keep every origin and each sample occurrence.

### Prediction examples — separate channels

Native: NOT_RUN when no provider ran
BA: affinity_nM = 50 (nM; lower is better)
EL: presentation_percentile = 0.8 (percent; lower)
IM: score = 0.7 (unitless; higher)

### Results and explanations

Native rank / somatic grade only where eligible
NOT_GRADED is not grade E.
External ranks are independent derived views.
Explain: candidate → all origins → all events.

Cohort examples: P1/S1 · P1/{S1,S2} · P1/{S1,S2} + P2/S3 — see the executable cohort guide.

**Synthetic scores demonstrate formats, not biological or clinical performance.**